

DASE7506 MP1 — Small Language Model Challenge Report

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1. Introduction

The goal of this project is to improve the provided small GPT language model and achieve a lower Bits Per Byte (BPB) score. The baseline model contains four Transformer blocks with a hidden size of 128 and four attention heads. It uses GELU in the feed-forward network and learned positional embeddings. My baseline achieved a validation BPB of **2.07109**.

I made two main changes to the baseline model:

- 1) Replace GELU with **SwiGLU**.
- 2) Replace learned positional embeddings with **RoPE**.

I also increased the training steps after selecting the improved architecture. The final model uses **SwiGLU + RoPE** and achieves a validation BPB of **1.69550** and a final test BPB of **1.72781**.

2. Method

2.1 SwiGLU

The baseline model uses a standard GELU feed-forward network. I replaced it with SwiGLU.

SwiGLU uses a gating mechanism:

$$\text{SwiGLU}(x) = \text{SiLU}(xW_1) \odot (xW_2)$$

where $\odot$ represents element-wise multiplication.

The main idea is that the gate allows the model to control which features are passed to the next layer. To keep the model size similar to the baseline, I used a hidden dimension of **344** for SwiGLU. The baseline has **1,088,256 parameters**, while the SwiGLU model has **1,093,056 parameters**. Therefore, the comparison is approximately parameter-matched.

2.2 RoPE

The baseline uses learned positional embeddings. I replaced them with **Rotary Positional Embeddings (RoPE)**.

RoPE adds positional information by rotating the query and key vectors inside the attention mechanism instead of adding a learned position vector to each token. Another advantage is that RoPE does not require the original positional embedding table.

The original position embedding contains:

$$256 \times 128 = 32,768 \text{ parameters.}$$

After using SwiGLU + RoPE, the final model has **1,060,288 parameters**.

3. Experimental Setup

All experiments use the provided WikiText-2 data and the fixed BPE tokenizer.

The main model settings are:

Setting	Value
Vocabulary size	2048
Context length	256
Hidden size	128
Attention heads	4
Transformer blocks	4
Random seed	17

I used validation BPB to compare models. Lower BPB means better performance.

The test set was only evaluated after the final model was selected.

4. Results

The main experimental results are shown below.

Model	Steps	Training Targets	Validation BPB
Baseline	1200	9.83M	2.07109
SwiGLU	1200	9.83M	2.00972
SwiGLU	2400	19.66M	1.82438
GELU + RoPE	2400	19.66M	1.77756
SwiGLU + RoPE	2400	19.66M	1.73986
SwiGLU + RoPE	4800	39.32M	1.69550

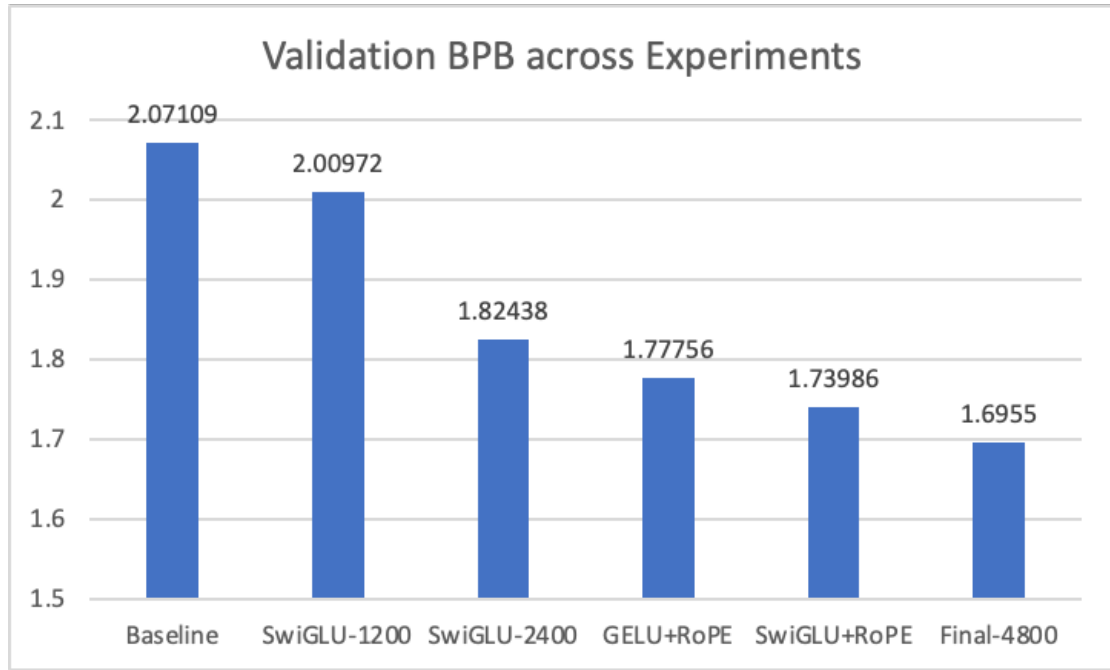


Figure 1. Validation BPB across different model configurations (lower is better).

As shown in Figure 1, each major modification reduced validation BPB. The final SwiGLU + RoPE model achieved the best validation result among the tested configurations.

4.1 SwiGLU Comparison

With the same 1,200 training steps, the baseline achieved **2.07109 BPB**, while the SwiGLU model achieved **2.00972 BPB**. This shows that SwiGLU improves the model even when the training budget is unchanged.

4.2 RoPE Comparison

At 2,400 steps, SwiGLU with learned positional embeddings achieved **1.82438 BPB**. After replacing the learned positional embeddings with RoPE, BPB decreased to **1.73986**.

Therefore, RoPE provided another clear improvement.

5. Ablation and Analysis

To check whether SwiGLU was still useful after adding RoPE, I compared two models using the same 2,400 training steps:

$$\text{GELU} + \text{RoPE} = 1.77756$$

$$\text{SwiGLU} + \text{RoPE} = 1.73986$$

SwiGLU + RoPE performed better. This suggests that SwiGLU contributes to the improvement even when RoPE is already used.

I then increased the SwiGLU + RoPE training budget from 2,400 to 4,800 steps. The validation BPB improved from **1.73986 to 1.69550**.

However, this experiment does not only change the number of training steps. The cosine learning-rate schedule also changes with the total number of steps. Therefore, this result should be considered an improvement from the overall training recipe rather than only from additional training steps.

There are also some limitations. Most experiments used only **seed 17**, so I did not measure the variation between different random seeds. In addition, I only tested a small number of model configurations because of the available training time.

6. Final Result and Conclusion

The final model uses:

$$\text{SwiGLU} + \text{RoPE} + \text{4,800 training steps}$$

It contains **1,060,288 parameters** and processes **39,321,600 training targets**.

After selecting the final model using the validation set, I froze the model and evaluated it on the complete test set.

The final result is:

Test BPB=1.727811

The validation BPB is **1.695495**.

Overall, both SwiGLU and RoPE improved the original baseline. SwiGLU improved the feed-forward network, while RoPE improved positional representation and also reduced the parameter count. Increasing the training budget provided an additional improvement.

The final model achieved substantially lower BPB than my baseline while remaining a small model.

References

- [1] N. Shazeer, “GLU Variants Improve Transformer,” 2020.
- [2] J. Su et al., “RoFormer: Enhanced Transformer with Rotary Position Embedding,” 2021.
- [3] A. Vaswani et al., “Attention Is All You Need,” 2017.

AI Assistance

I used ChatGPT (OpenAI) for explanations, experiment planning, debugging assistance, result analysis, and report writing assistance. I reviewed the implementation and ran and verified all experiments locally.